

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Comparative Analysis

Course Name: Computer Networks

Course Code: CSA07

Course Outcome: CO4 — Design network topologies (star, mesh, bus, and hybrid) using networking devices, transmission media, and cabling standards

Assessment Weightage: 30%

Total Marks: 25 (5 Questions)

Code of Conduct

I, C KARTHIK REDDY(192425157), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Karthik Reddy

Comparative Analysis – Answers

Q1. Star vs Bus Topology — Cost, Reliability, and Performance

Cost Comparison

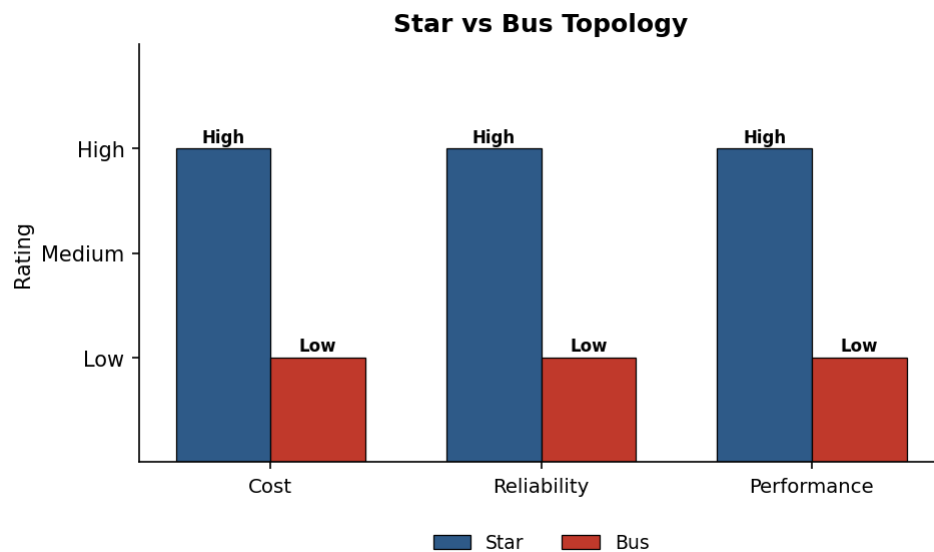
Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install.

Reliability Comparison

In terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network.

Performance Comparison

Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.



Q2. Mesh vs Star Topology — Fault Tolerance

Mesh Topology Fault Tolerance

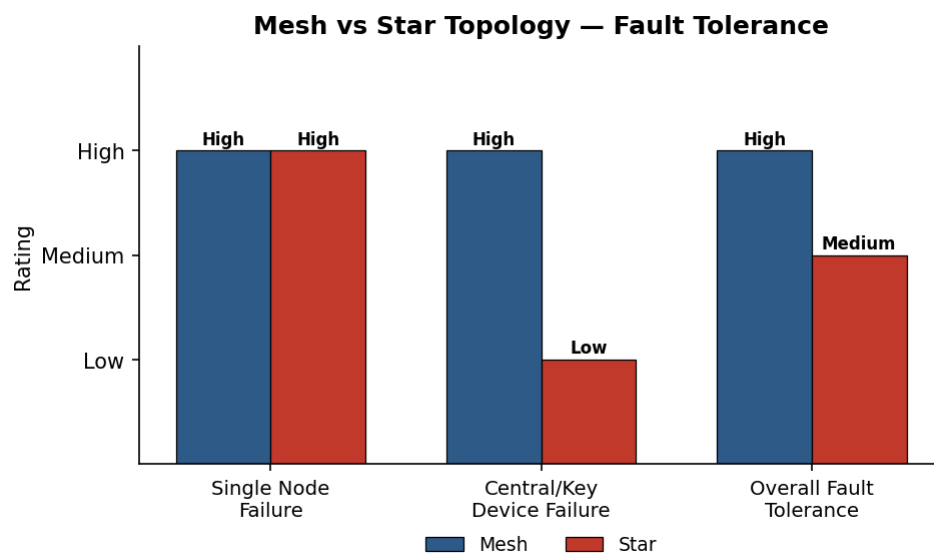
Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication.

Star Topology Fault Tolerance

In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning.

Conclusion

Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.



Q3. Bus vs Mesh Topology — Installation Complexity

Bus Topology Installation

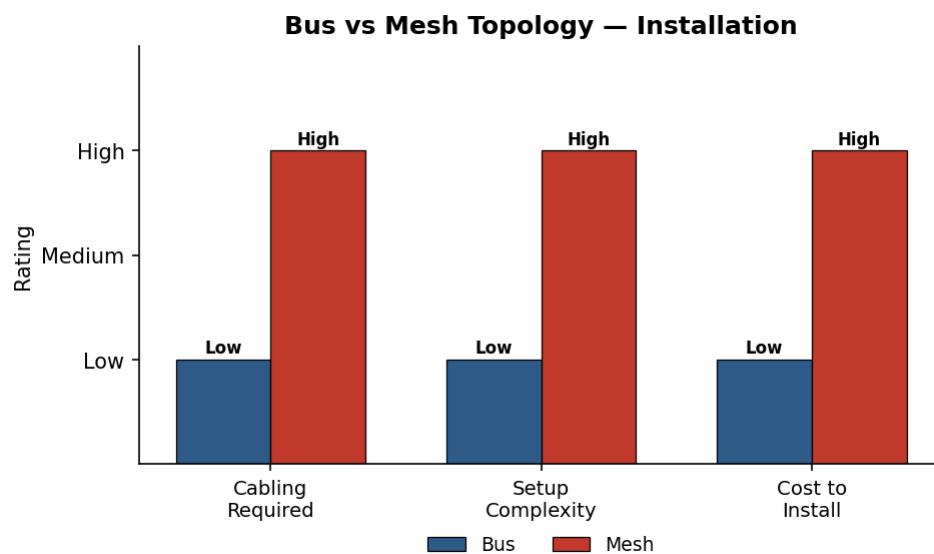
Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks.

Mesh Topology Installation

On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports.

Scalability of Complexity

This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.



Q4. Hybrid vs Star Topology — Scalability

Hybrid Topology Scalability

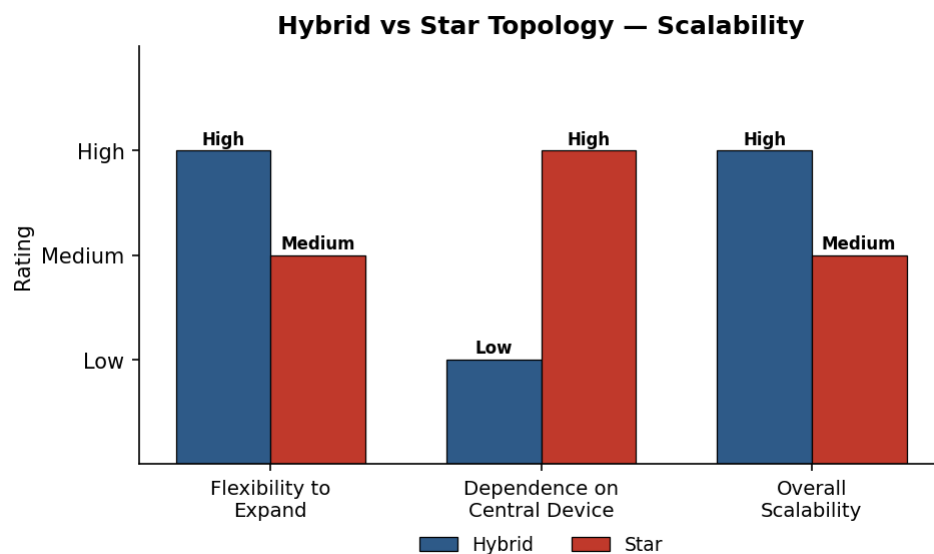
Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure.

Star Topology Scalability

In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware.

Conclusion

Thus, hybrid topology provides greater scalability than star topology.



Q5. Maintenance Comparison — Star, Mesh, Bus, and Hybrid

Star Topology Maintenance

In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable.

Mesh Topology Maintenance

Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming.

Bus Topology Maintenance

Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate.

Hybrid Topology Maintenance

Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Ease of Maintenance by Topology

